Designing a Universal Information Language

Oscar Bender-Stone

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DRAFT

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Background

- Information Management: studies how to **store** and **organize** data

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  - Example: Amazon (product suggestions)[1]

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- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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Existing Solutions

- **Resource Description Language (RDF)**
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - − No built-in logic engine
- **Labeled Property Graphs (LPGs)**
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - − Not standardized
- **Cyc**
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - − Hard-coded, proprietary rules

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